Describing a numerical variable

MATH 213

Key Features

When analyzing a numerical variable, we pay close attention and try to describe the following features of the distribution:

- Who / what / when / where
- Shape
- Center
- Spread
- Outliers, if any
- Other interesting features

Who / What / Where / When

- This one is self-explanatory!
- Situate the reader – give them the context they need to understand the details you are going to provide next.

Shape: Commonly observed shapes of distributions

Skewness

right skew

left skew

symmetric

Modality

unimodal

bimodal

multimodal

uniform

Sample language for describing shape

- The distribution of ages is **right-skewed** meaning that more of the Oscar-winners are in their 20’s and 30’s and a smaller number are older than that.
- (Use analogous language for a **left-skewed** distribution.)
- (A made-up example.) Oscar-winners’ ages are distributed **symmetrically**, meaning that there were about as many of the actors below 40 years of age as there were above age 40. (“40 years” – replace this with your median.)

Center: Measures of Center

The **mean** (what people mean when they say the “average”) is calculated by adding up the observations of the numerical variable and dividing by however many there are.

$$\frac{x_1 + x_2 + \cdots + x_n}{n},$$

The **median** is the value that splits the data in half when ordered in ascending order.

E.g.:

$$0, 1, 2, 3, 4$$

If there are an even number of observations, then the median is the average of the two values in the middle.

$$0, 1, \underline{2, 3}, 4, 5 \rightarrow \frac{2+3}{2} = 2.5$$

The median is the midpoint of the data.

Tip: Medians give a better sense of the center for a skewed distribution. Means are better when describing symmetric distributions.

Language for describing center

A **typical** actor was approximately 40 years of age.

Oscar winners **tended to be around** 40 years old.

Can use this language for means or medians.

Don’t use the word average unless you use the mean.

Remember: medians are better for skewed distributions. Means are better for symmetric distributions.

Spread: Measures of spread

Standard Deviation

Has a complicated-looking formula:

$$\sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

Standard deviation measures roughly how far away a typical observation is from the mean. (Memorize this.)

Variance

The variance is another measure of spread. It is calculated as the square of the standard deviation.

Quantiles

In a moment, we will learn how to use R to calculate the 0.025, 0.25, 0.75, and 0.975 quantiles.

Example: the 0.75 quantile = the 75th percentile, i.e. 75% of the data fall below this value. The others have analogous definitions.

This means that we can talk about the **middle 50% of the data** - between the 0.25 and 0.75 quantile values, and the **middle 95% of the data** - between the 0.025 and 0.975 quantiles.

Language for describing spread

In your descriptions, delete the pink text and fill in actual values for std deviation, quantiles, etc.

Example incorporating standard deviation:

- A typical actor was around 40 years old, **give or take** *(standard deviation)* **years.**

Example for incorporating quantiles

- The **middle 50%** of actors were between *(0.25 quantile and 0.75 quantile)* years old. The **middle 95%** were between *(0.025 quantile and 0.975 quantile)* years old.

or

- **Most of** the actors (the middle 50%) were between *(0.25 quantile and 0.75 quantile)* years old and **almost all** (the middle 95%) were between *(0.025 quantile and 0.975 quantile)* years of age.

Outliers

- Informal definition: Outliers are values in the dataset that appear extreme relative to the rest of the data.
- Pay close attention to outliers - do not automatically ignore them. Try to investigate them and find reasons why they exist.
- In our descriptions, highlight outliers by mentioning them (their values) specifically.

Example language: There are outliers to the general trend, such as Anthony Hopkins winning Best Actor at 83 years old in 2021 or Jessica Tandy who was 80 when she won for Best Actress in 1990.